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Date: 5/29/2025

Class: 6

MyDesign Portfolio

Element J Initial Template

Element J: Documentation of Evaluation

Stakeholder Evaluation #1

- Name of evaluator: Homeowner
- Justification for how this evaluator is a **stakeholder**: Homeowners are a major stakeholder in our project because we are designing a house with fire protection systems which could potentially save a homeowner's life in the case of a fire.
- As residents of a home, we evaluated our own design and determined that the furniture placement could be improved to be more realistic (not having a bed in the living room). We also thought that the sprinkler system would have to be hidden because in a normal house, the entire system would not be visible.

Stakeholder Evaluation #2

- Name of evaluator: Fire Department
- Justification for how this evaluator is a **stakeholder**: The Fire Department is a stakeholder because they are affected by the problem of fire-related accidents. They are responsible for protecting people who are affected by failed fire detection and suppression systems. Therefore, they need to have a say in our design.
- We did not meet this stakeholder. However, we looked at a report of fire-related deaths and injuries in Maryland and determined that this was a rising problem which affected the Fire Department. We knew that our working detection system would benefit the fire department because they would have the opportunity to save people immediately when a fire is detected. However, our suppression system, which failed, would affect the fire department negatively because it would put a risk to their lives. Therefore, we knew that the suppression system (sprinklers) would need to be redesigned in the future.

Field Expert Evaluation #1

- Name of evaluator: Dr. Jim Milke

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- Justification for how this evaluator is a **demonstrably qualified field expert** (professionals who spend at least a portion of their time working in a field directly related to your project): Dr. Milke is a field expert because he works as a professor in the University of Maryland's Fire Protection Engineering Department. He holds a B.S. in Fire Protection Engineering and a M.S. in Mechanical Engineering, as well as a Ph.D. in Aerospace Engineering, all from the University of Maryland, whose engineering department organized the Burn Day challenge. Therefore, Dr. Milke is knowledgeable in the field of our project and can provide feedback and information regarding fire detection and suppression systems.
- We had an in-person conversation with this field expert. He said that our fire detection system works well because it began beeping loudly as soon as the fire was lit. However, he said much more water was needed for our suppression system in order to put out the fire. Additionally, we needed to secure the sprinkler "bags" to the ceiling so that they would not fall off.

Synthesis of the Project External Evaluation

The external evaluation means that our design has the potential to work very well, but a few changes need to be made to it. For example, our design would have worked completely if the suppression system was redesigned and the house was rebuilt. From this evaluation, I am taking away that we need to talk to our stakeholder evaluators in order to receive expert feedback that will improve our design before we test it. Additionally, we need more field experts so that they can critique our design in ways that we would not know using their knowledge in the field. Next time, when working on a design, my team needs to arrange meetings with the stakeholders and also with the field experts in order to develop working systems that will solve our problem.